

TRADERS WORLD AI

Project Execution & Technical Summary Report | Founder Build

1. Executive Summary

Traders World AI is a complete, production-grade **AI Trading Operating System**. Unlike generic trading bots or signal providers, Traders World AI is designed as a personal AI mentor that deeply learns one specific trading methodology (Smart Money Concepts — HTF bias, A+ zones, liquidity, ChoCH/BOS structure shifts, confirmation candles, risk management, and psychology) and assists traders with unwavering consistency.

Core Philosophy: The AI never predicts market direction. It only evaluates and explains *why* a trade is valid or invalid according to documented strategy rules.

Component	Technology / Spec	Status
Frontend App	Next.js 14 (App Router) + TypeScript + Tailwind CSS	Completed & Live
Backend Engine	FastAPI + Python 3.11 + Async Engine	Completed & Live
AI Brain	Google Gemini 1.5 Flash (1M tokens/day FREE)	Configured & Active
Voice Copilot	Browser Native Web Speech API (STT & TTS)	Completed & Free
RAG Engine	ChromaDB + BM25 Hybrid Search + Gemini Reranker	Completed
Database Schema	SQLAlchemy 2.0 Async (9 Entities)	Initialized & Migrated
Docker Infra	Docker Compose (Next.js + FastAPI + ChromaDB)	Completed
GitHub Repo	shubhamhulsure11-lang/Traders-World-ai	Pushed & Live

2. System Architecture & 5-Stage AI Pipeline

The core backend features an advanced **5-Stage AI Orchestrator Pipeline** that processes every request transparently without letting the LLM hallucinate or guess strategy rules:

- **Stage 1 — Intent Detection:** Fast Gemini Flash call classifies message intent (live_analysis, teaching, checklist, journal_review, psychology, etc.) and assigns AI role.
- **Stage 2 — Knowledge Retrieval (RAG):** Executes hybrid BM25 + Vector search over ChromaDB with Reciprocal Rank Fusion (RRF) and Gemini Flash reranking.
- **Stage 3 — Strategy Validation:** Strategy engine evaluates live market context against 8 SMC rules (HTF bias, A+ zone, liquidity, ChoCH/BOS, confirmation candle, risk <= 2%).
- **Stage 4 — Response Generation:** Prompt builder assembles 8-layer prompt context and streams output adhering to rigid Observation / Evidence / Rule / Explanation / Recommendation structure.
- **Stage 5 — Async Memory Update:** Non-blocking background task records turn into Layer 2 Conversation Memory and Layer 3 Trading Session Memory tables.

3. 4-Layer Memory System Architecture

Memory Layer	Scope & Storage	Persistence & Rule
Layer 1: Working Memory	Single Request (In-Memory Dict)	Never persisted. Discarded after response.
Layer 2: Conversation	Current Chat Session (DB table)	24-hour TTL active memory, archived to DB.
Layer 3: Trading Session	Daily Trading Session (DB table)	Permanent DB record of daily bias & setups.
Layer 4: Long-Term Knowledge	ChromaDB + Markdown Repository	Permanent strategy brain. Manual update only.

4. Database Schema (SQLAlchemy 2.0 Async)

The application relies on 9 fully typed SQLAlchemy models in backend/database/models.py:

- **User:** User profiles, role (founder/premium/standard), preferences.
- **Conversation:** Chat sessions with mode (text/voice), token counters, timestamps.
- **Message:** Full history with intent, citations, retrieval score, latency, and token metrics.
- **TradingSession:** Daily session notes, symbol, session type, HTF bias, mistake tracker.
- **Trade:** Complete trade log (entry, SL, TP, risk %, R-multiple, setup quality, rules broken/followed).
- **TradeScreenshot:** Linked before/after/annotated chart screenshots.
- **Backtest:** Historical backtest statistics (sample size, win rate, avg RR, profit factor, max DD).
- **KnowledgeDocument:** Metadata tracking for strategy markdown files and chunk index counts.
- **AIEvaluationLog:** Automated regression testing logs, hallucination flags, human ratings.

5. Frontend & Workspace Features

Built with **Next.js 14 App Router**, Tailwind CSS, and Lucide icons, providing a dark-mode premium interface:

- **Dashboard Workspace:** Side-by-side view with TradingView chart widget, 8-rule A+ Pre-Trade Checklist, and live RAG AI Copilot panel.
- **Voice Copilot Page:** Native Web Speech API STT and TTS voice integration for hands-free trading mentoring.
- **Trade Journal Page:** Structured trade log interface with rule tracking and setup quality tagging.
- **Backtest & Analytics Page:** Statistical dashboards tracking win rate, profit factor, RR, and psychological metrics (FOMO count, discipline score).
- **Knowledge Base Page:** Search engine and one-click re-indexing trigger for ChromaDB vector store.

6. Deployment & GitHub Status

GitHub Repository: shubhamhulsure11-lang/Traders-World-ai

Commit Status: All 97 codebase files committed and force-updated to main branch.

Local Live Servers:

- Frontend Dashboard: <http://localhost:3000>
 - Backend API & Docs: <http://localhost:8000/docs>
 - Health Check: <http://localhost:8000/health> (Status: Operational)
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